

A Rotation & Scale Invariant CNN Model to Detect Humans in Disaster Area from Aerial View

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Abstract—Human detection in aerial images is incredibly challenging because of the characteristics aerial images possess. One of the main applications of human detection in aerial images is for search and rescue operations after a disaster. An efficient human detection approach will hugely optimize the search and rescue operation, which helps reduce the loss of lives due to disasters. A custom dataset has been made for this study using an Unmanned Aerial Vehicle (UAV) or drone. The dataset is made to simulate real-life disaster scenarios. Various types of preprocessing are done on the dataset considering the wide range of aerial images, i.e., different scales, orientations, and postures of the human body cover different areas in aerial images. A new convolutional neural network (CNN) based model has been proposed for classification in this paper. The model has been made both scaling and rotation invariant. A comparative study has been done among 12 pretrained models and compared to the proposed model. You Only Look Once (YOLOv3) has been used for transfer learning for the task of localization. The proposed classification model has shown satisfactory results with accuracy as high as 94.18%.

Index Terms—Human detection, CNN, image classification and localization, aerial images.

I. INTRODUCTION

After a disaster strikes, the first hours are highly crucial. Proper rescue missions during this time can help save many lives and mitigate the aftermath of a disaster. However, it is hazardous to immediately send a rescue team consisting of human beings. Following the aftermath of the disaster, it is not always favorable to transport the rescue team to the disaster site. It also imposes a threat to the lives of the rescuers. Moreover, the rescue team cannot just go and start helping the injured people. They need to locate the injured people and document their locations. This search operation of a rescue mission can be time-consuming and painstaking. There might be circumstances and places where the rescue team cannot go and search for the affected people. Therefore, a drone can be used to locate the injured people from aerial images to optimize the rescue operation. Due to transportation limitations, it is not always possible for the rescuers to carry all the necessary equipment or aid kits in abundant quantities. Therefore, the rescue team may face a shortage of aid kits during their rescue missions. Once the locations of the victims are documented through drones and the severity of the circumstances is assessed, necessary equipment and aid kits can be sent with the rescue team in proper quantity.

Using aerial images for human detection has different challenges. The size of the human body in aerial images varies for many reasons. First, the variation in altitude and orientation of UAVs can cause variations in human body sizes and shapes in aerial images. The human body takes up a small percentage of the entire aerial image. In other words, the human body consists of a small number of pixels in aerial images. In disaster-struck areas, the image will contain different background clutters. The system has to distinguish the human body from the background clutters.

The injured and affected people can be found in different situations, positions, and orientations in disaster-struck areas. The affected human body can be found sitting, standing, lying down, or in different positions, causing human body deformities in aerial images. The orientation of the affected human body also varies to different degrees. The body is not always seen head up in aerial images. Moreover, the debris resulting from the disaster can occlude the human body. The system should be able to detect the affected human bodies in all possible ways mentioned above. The contribution of the paper is summarised as follows:

- 1) The dataset used in this paper is a customized dataset that has been created using UAV.
- 2) The proposed CNN model has achieved higher accuracy on the custom dataset while decreasing the model's parameters to half compared to the pretrained model like VGG-16 or VGG-19.
- 3) The proposed model's performance on the custom dataset has been compared to 12 other pretrained models' performances on the custom dataset.

II. LITERATURE REVIEW

Human detection has been an area of interest of many researchers. Basically, human detection is done in three ways: deep learning based method (usually transfer learning approach), feature extraction based methods, and image processing based methods.

Zacharie et al. [1] performed image conversion from Red, Green, Blue to Hue, Saturation and Value. For skin color recognition, they performed threshold processing on the converted image. Their method can detect full human bodies as well as partially occluded ones. If the color of anything near the environment falls in the range of human body skin color,

it may get falsely detected. Again, the particles of debris may cover the skin and change the skin color. In this scenario, detection of the human body using skin color would not be possible.

Rudol and Doherty [2] used an algorithm that takes two input images and starts analyzing the thermal image in order to detect human-temperature silhouettes. After detecting a human body shape, corresponding regions are calculated in the colored-image. One of the corresponding pixels in the latter was identified as the center of the object for object detection. They used Leinhardt's extended set of available features. The classifier used was a cascade of boosted classifiers that work with Haar-like features. They adjusted the parameters of the classifier to minimize false negative cases.

Portmann et al. [3] proposed a pipelined method that involves background subtraction and a particle filter guided detector. A predefined threshold value is set for background pixels. Each individual pixel of an image is labeled as background if the pixel falls within the mentioned threshold value, thus performing background subtraction. For detectors, they have used Histograms of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Body Part based Detector (BPD). As their final step, they used trackers for detection and then evaluated their performance.

Wang et al. [4] used HOG and LBP method on the INRIA dataset. HOG contributes to storing the information of the local shape, which is combined with LBP, which stores the texture information and helps to remove noises based on the uniformity of pattern. Upon acquiring the responses of the HOG feature, an occlusion probability map is produced for each ambiguous scanning window. The part-based detector is then used on the portions that are not occluded to get the final classification for the present scanning window. This method gives the best performance on the INRIA dataset while handling partial occlusion. This method is only applied to pedestrian classification. It may not handle the deformation of the human body. Applying this to aerial images may not give satisfactory results.

Oreifej et al. [5] proposed an algorithm to detect the human body from aerial images with low-quality definitions. All the input images and query images are preprocessed using the procedures: 1. Human Detection 2. Blob Extraction & 3. Alignment. They trained a support vector machine classifier on the basis of HOG descriptor for detection. They chose to use Dalal and Triggs' algorithm [6] for aerial images.

Aïdahoul et al. [6] applied three different kinds of deep models. For supervised learning, Stochastic Gradient Descent algorithm was employed to train the Supervised-CNN network. For unsupervised learning, Hierarchical Extreme Learning Machine is used. For pretrained CNN model, they have used AlexNet [7]. For the optical flow stabilization video frames are utilised as input. They compared three different deep models on detecting humans from aerial videos in real-time and all of them gave above 90% accuracy. They found the best result on pretrained CNN model. The methods are only applied on five different activities of humans. Partial occlusion

and deformation of human body is not handled.

Vasic and Papic [8] proposed a new multimodal approach to detect human body from aerial images. Their proposed multimodal model is a combination of two different CNN architectures for two stages. They have improved the results in the classification phase by using the contextual data regarding the potential region's surroundings. They have successfully achieved better results on the HERIDAL dataset compared to other recent models. They have obtained a precision of 68.89% and a recall of 94.65%.

Chang et al. [9] used deep learning based models for pedestrian detection from aerial images while integrating two methods in preprocessing steps. The first step is image partition, which resolves the issue of small pedestrian sizes in aerial images. They performed a vanishing point transformation on the sub-images for their second step. Applying this transformation, the human body orientation would come to an upright position regardless of their original orientation. Finally, after merging these two steps, the image is sent to YOLOv2 for pedestrian detection.

III. METHODOLOGY

A. Dataset Preparation

A new dataset is created for the task of detecting humans in disaster areas from an aerial view. Pictures are captured using a camera on a drone from high altitude and a mobile camera from the rooftop of buildings to get a high range of variety of aerial views of humans. Some people volunteered in different postures to impose humans stuck in a disaster area. Pictures are taken at different angles and altitudes and different combinations of pose and position. Partially occluded human bodies are also captured. Some images are collected from online also.

The purpose of preprocessing is to enrich the dataset, enabling deep neural networks to access different possible cases for feature extraction. The dataset for this paper has been augmented in the following way: First, three scaled versions of images were used - large, medium, and small. The human coverage percentage in the entire image has defined these three different scales. The large, medium, and small-scale images contain human coverage of approximately 30%, 15%, and 5% or less, respectively. Next, four different rotations (45, 90, 135, and 220 degrees) were applied to different portion of scaled images to capture the different angles of human positions. The rotations have been performed with border copy mode. All the images are then resized into the standard shape of $224 \times 224 \times 3$ for the model. Images are resized using enter area interpolation method as they are down-sampled. In this approach, it resamples utilizing the pixel area relation. It is a preferable approach for picture reduction because it is designed for decimation and can avoid the aliasing effect. However, when the picture is zoomed, the technique behaves similarly to the INTER NEAREST method, which uses nearest-neighbor interpolation.

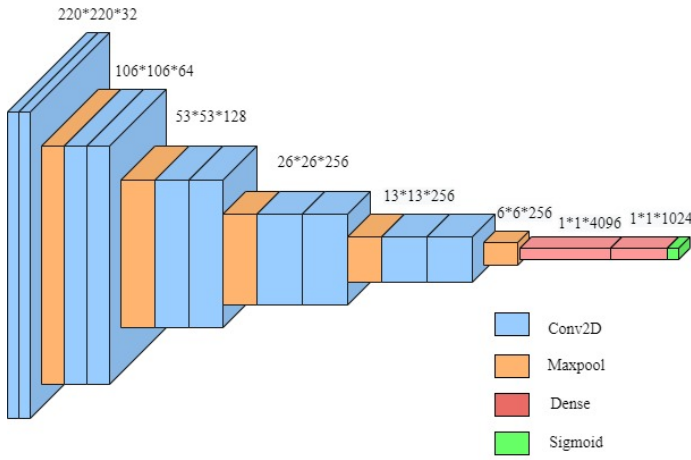


Fig. 1: Architecture of the proposed CNN model.

B. Proposed CNN Model

A new CNN [10] model is proposed for classification by using trial and error method. As image dataset is used in this research, CNN is used to automate the feature extraction process. Multiple iterations allow CNNs to learn an object's features, negating the need for labor-intensive human feature engineering operations. As spatial information is vital for image dataset, 2D convolution is chosen for this work. Transfer learning is applied on the custom dataset to perform a comparative study between the proposed CNN model with other pretrained CNN models. Localization is done on the positively classified images by YOLOv3 [11].

The proposed model consists of 13 weighted layers with 5 convolution blocks. Each convolution block has two convolution layers and one max pooling layer. A batch normalizing layer is applied immediately following each convolution layer. For all five blocks kernel size of 5×5 is used. The first three blocks have 32, 64, and 128 kernels respectively. The number of kernels in both fourth and fifth blocks is 256. Two dense layers with 4096 and 1024 neurons are added after the convolution blocks. A dropout layer with parameter 0.25 is added to regularise the network. In all hidden layers Relu [12] activation function is used. Sigmoid [13] activation function is utilized for classifying human and non human classes. The model consists of around 48 million parameters. The pictorial view of architecture of the proposed model is shown in Fig.1.

The details about the proposed CNN model's architecture is documented in tabular format in Table I. The model is compiled with binary cross entropy as the loss function and accuracy metrics and Adam optimizer [14]. In order to regularize the model, the training will be stopped if the validation loss does not go down in 10 successive epochs as a process of early stopping. For both the training and validation datasets, the initial learning rate is 10^{-4} and the minimum learning rate is 5×10^{-7} . For both training and validation sets, 32 batch size is employed to train the model. After classifying positive and negative images, YOLOv3 [11] is applied on the positive images to localize the human bodies in the aerial image. A

TABLE I: Tabular representation of proposed model's architecture

Layer	Filters	Size	Output
Convolutional	32	5×5	220×220
Convolutional	32	5×5	220×220
Max pooling		2×2	110×110
Convolutional	64	5×5	106×106
Convolutional	64	5×5	106×106
Max pooling		2×2	53×53
Convolutional	128	5×5	53×53
Convolutional	128	5×5	53×53
Max pooling		2×2	26×26
Convolutional	256	5×5	26×26
Convolutional	256	5×5	26×26
Max pooling		2×2	13×13
Convolutional	256	5×5	13×13
Convolutional	256	5×5	13×13
Max pooling		2×2	6×6
Flatten			9216
Fully Connected			4096
Fully Connected			1024
Sigmoid			1

boundary box is predicted for each human bodies identified by YOLOv3.

C. Pre-trained Models

Transfer learning is applied to the customized dataset to observe how well these models can classify humans in disaster areas from an aerial view. Transfer learning saves a lot of resources and is efficient in building new models. Deep learning models available as Keras [15] applications are trained on ImageNet. These models have initialized weights. The WordNet hierarchy estimates that there are over 14 million annotated images in the ImageNet dataset. Since 2010, the dataset has been utilized in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), a benchmark for picture classification and object recognition. The pre-trained models that are used on the dataset are: ResNetv2 [16] (ResNet50v2, ResNet101v2, ResNet152v2), VGG [17] (VGG16, VGG19), MobileNet (MobileNetv1 [18], MobileNetv2 [19]), DenseNet [20] (DenseNet121, DenseNet169, DenseNet201), Inceptionv3 [21] and Xception [22].

Fig. 2 shows how transfer learning is used to train on the custom dataset. The base layer takes input images in $224 \times 224 \times 3$ shape. The last layer to classify 1000 classes of ImageNet is removed. A flatten and dropout layer of 0.5 are stacked with the base layer. To classify human and non-human classes, in the last layer sigmoid activation function is applied.

IV. RESULT ANALYSIS

A. Dataset Description

There are two different classes in the dataset: Human and Non-human. Dataset is created by taking pictures using drone from high altitude and mobile device from the rooftop. Some images are collected from online. The original dataset consists of 1000 images having 500 images in each class. Dataset is split into three sections for training, validation, and testing in ratio of 70%, 15%, and 15%, respectively. Augmentation

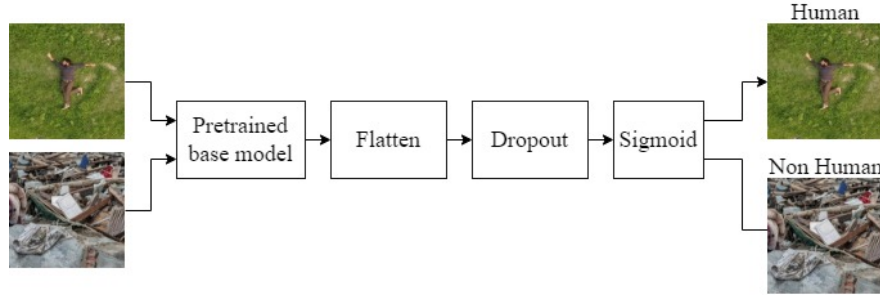


Fig. 2: Diagram to train the pre-trained models.

TABLE II: Dataset description after augmentation

Images	Training set	Validation set	Test set	Total images
Positive	2000	375	75	2450
Negative	2000	375	75	2450
Total	4000	750	150	4900

is applied to the training and validation set to increase the volume of the dataset and to let the models learn many versions of human images by applying different rotations in different portions of image dataset, scaling and shifting so that the models can be scale, shift and rotation invariant. After applying augmentation, the number of images in each class is documented in Table II.

B. Evaluation Metrics

The classification performance of the models is evaluated based on four statistical measures: accuracy, recall, precision, and F1 score. The mentioned statistical measures are calculated using the four terms from the confusion matrix: True Positive, True Negative, False Positive, and False Negative.

- The accuracy score represents the proportion of successfully predicted data points to all data points.
- Precision score measures the ratio of the number of correctly predicted positive data points to all other predicted positive data points.
- Recall score measures the proportion of the number of accurately anticipated positive data points compared to the total number of actual positive data points.
- The F1 score denotes the weighted harmonic mean of the recall and precision of the test.

C. Experimental Results

To evaluate the performance of the models, 15 different versions of test sets are applied to them. There are three scaled versions (small, medium, and large) of the test set according to the area covered by human body in the aerial images. Four rotations: 45, 90, 120, and 270 degrees are applied on each scaled test sets to test the reliability of the model in rotated and scaled aerial images. Evaluation metrics are shown in Table III for each test set. On average of all three scaled images test sets, the proposed model has achieved an outstanding performance with 96.44% accuracy, 97.28%

TABLE III: Result of proposed model

Test sets		Accuracy	Precision	Recall	F1 score
Rotation	Scale				
0	Small	96.67	97.30	96.00	96.64
	Medium	96.67	97.30	96.00	96.64
	Large	96.00	97.26	94.67	95.95
45	Small	96.67	98.61	94.67	96.60
	Medium	98.00	98.65	97.33	97.99
	Large	96.67	98.61	94.67	96.60
90	Small	88.67	96.77	80.00	87.59
	Medium	95.33	97.22	93.33	95.24
	Large	96.67	97.30	96.00	96.64
120	Small	86.00	95.00	76.00	84.44
	Medium	96.00	96.00	96.00	96.00
	Large	95.33	95.95	94.67	95.30
270	Small	87.33	93.75	80.00	86.33
	Medium	94.00	94.59	93.33	93.96
	Large	92.67	94.44	90.67	92.52
Average		94.18	96.58	91.56	93.90

precision, 95.56% recall, and 96.41% f1 score. On the 12 rotated test sets it has obtained 93.61% average accuracy with 96.41% precision, 90.56% recall and 93.27% f1 score. On average of 15 evaluations, the proposed model gives 94.18% accuracy.

Fig. 3(a), (b) and (c) shows the output result on scaled images in 3 categories: small, medium and large respectively and 3(d), (e) and (f) shows 45, 120 and 270 degrees rotated image output of same image. Fig. 4 shows the output of YOLOv3 on positively classified images by the proposed CNN model. Fig. 4(a), (b) and (c) have some human that have not been detected by the system. It has been seen that this occurs where the presence of the human body is less than 1% of the total image size. Such cases make it difficult for the human eyes to locate, so it is even more challenging for the model to extract the features and detect the presence of a human body on such small scales. The systems detects partially occluded cases as shown in Fig. 4(e).

A comparative study among the pre-trained models and the proposed model is also done. Average of the all 15 evaluation of 13 models are documented in Table IV. It shows that the proposed model has performed the best with the highest accuracy, precision and F1 score with the value of 94.18%, 96.58%, and 93.90% respectively. The highest recall is obtained by ResNet152v2 with accuracy score of

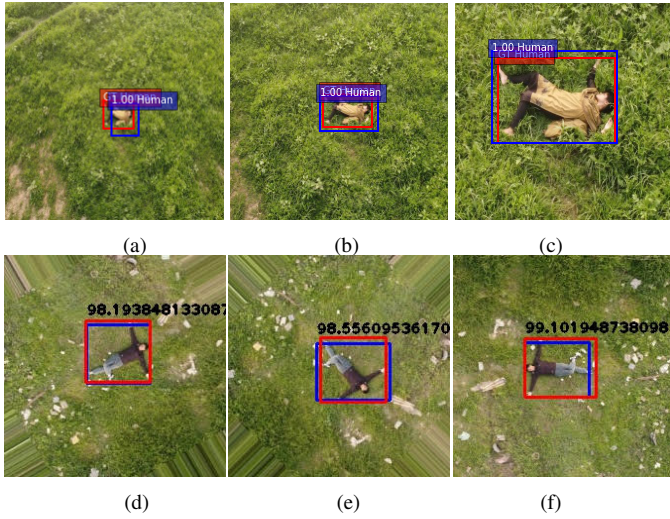


Fig. 3: Images (a), (b) and (c) shows small, medium and large scaled version of the same image. Images (d), (e) and (f) shows 45, 120 and 270 degrees rotated version of the same image. Here, red boxes indicate ground truth box and blue boxes indicate predicted box.

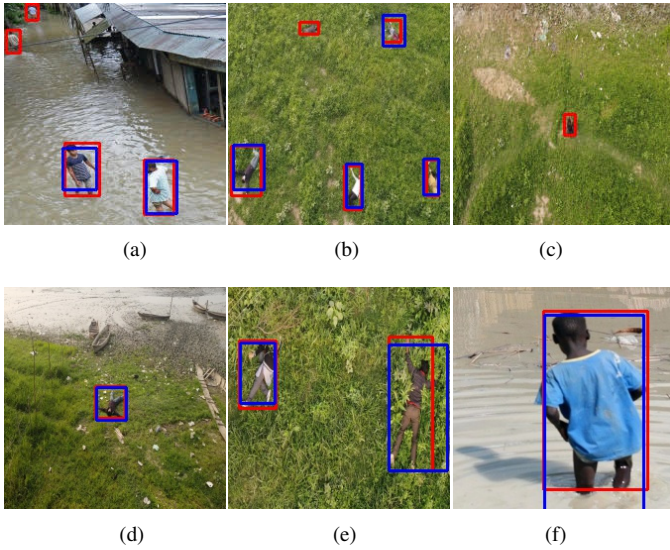


Fig. 4: Some sample output of the positive images. Here, red boxes indicate ground truth box and blue boxes indicate predicted boxes.

TABLE IV: Comparison result analysis

Models	Accuracy	Precision	Recall	F1 score
Proposed Model	94.18	96.58	91.56	93.90
ResNet50v2	88.40	88.68	88.27	88.09
ResNet101v2	87.96	88.83	87.38	87.58
ResNet152v2	89.82	88.39	91.91	89.91
VGG16	88.89	89.15	89.51	88.92
VGG19	90.36	93.31	88.44	90.08
MobileNetv1	78.71	82.08	73.16	76.42
MobileNetv2	81.11	84.99	75.56	79.37
DenseNet121	82.44	81.19	84.89	82.55
DenseNet169	78.58	81.60	72.36	75.06
DenseNet201	80.53	85.73	73.07	77.25
InceptionV3	71.91	69.29	78.76	73.57
Xception	75.87	76.40	73.33	74.11

89.82%. VGG19 has achieved the second highest accuracy with 90.36%.

V. CONCLUSION

In a world where human beings coexist with disasters, human detection in aerial images plays a vital role in search and rescue missions and more. The nature of aerial images makes the task of human detection more challenging. A customized model has been proposed for human classification, which handles single and multiple objects in aerial images. The model has been built as a translation (scaling and rotation) invariant. The accuracy score for the proposed model is 94.18% which has been the best accuracy score so far among the pre-trained models used for comparison. Transfer learning has been performed in 12 deep CNN models (ResNetv2, VGG19, VGG 16, DenseNet, MobileNet, Inceptionv3, Xception) for classification whereon, YOLOv3 is used to detect the boundary box. In future work, the dataset can be increased and diversified so that the classifier becomes more robust. The system struggled to perform well on images where the ratio of the human body present in the image is minuscule (less than 1% human body coverage) compared to the entire aerial image. In such cases it becomes difficult for the model to extract the features and detect them. Moreover, the scope of this paper limits its work to static images. It can be broadened to work with dynamic real-time video footage for future work.

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